

Linear Algebra (MTS203)

Final Exam, (180 mins — 60 Marks)

Class no. 40241 (Spring 2024)

Exam Date - May 27, 2024

Note: *Only attempt questions amounting to an exact total of 60 marks.*

Question 1: (3 Marks)

Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array} \right]$$

For what values of β will the system be inconsistent?

Question 2: (2 Marks)

Show that if A is a symmetric nonsingular matrix then A^{-1} is also symmetric.

Question 3:(2 Marks)

Show that if A is row equivalent to I and $AB = AC$, then B must equal C .

Question 4:(1 Mark)

Prove or disprove: If A is row equivalent to both B and C , then A is row equivalent to $B + C$.

Question 5:(5 Marks)

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

where A_{11} is a $k \times k$ nonsingular matrix. Show that A can be factored into a product

$$\begin{bmatrix} I & O \\ B & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ O & C \end{bmatrix}$$

Find an expression for B and C in terms of A_{ij} and/or their inverses (if they exist).

Question 6: (5 Marks)

Write a proof: If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Question 7: (5 Marks)

Consider the 3×3 Vandermonde matrix

$$\begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}$$

Show that $\det(V) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$. What conditions must the scalar x_1, x_2 , and x_3 satisfy in order for V to be nonsingular?

Question 8: (3 Marks)

Suppose that Q is a matrix with the property $Q^{-1} = Q^T$. Show that

$$q_{ij} = \frac{Q_{ij}}{\det(Q)}$$

Question 9: (7 marks)

Let R denote the set of real numbers. Define scalar multiplication by

$$\alpha x = \alpha \cdot x. \quad (\text{the usual multiplication of real numbers})$$

and define vector addition, denoted $\oplus$, by

$$x \oplus y = \max(x, y) \quad (\text{the maximum of the two numbers})$$

Is R a vector space with these operations? Prove your answer.

Question 10: (7 marks)

State whether the following are subspace (or not a subspace) of $\mathbb{R}^{2 \times 2}$.

- (a) The set of all 2×2 diagonal matrices.
- (b) The set of all 2×2 triangular matrices.
- (c) The set of all 2×2 lower triangular matrices.
- (d) The set of all 2×2 matrices A such that $a_{12} = 1$.
- (e) The set of all 2×2 matrices B such that $b_{11} = 0$.
- (f) The set of all symmetric 2×2 matrices.
- (g) The set of all singular 2×2 matrices.

Question 11: (3 marks)

State whether the following are subspace (or not a subspace) of P_4 .

- (a) The set of polynomials in P_4 of even degree.
- (b) The set of all polynomials in P_4 having the property $p(0) = 0$.
- (c) The set of all polynomials in P_4 , having at least one real root.

Question 12: (3 marks)

Let A be a fixed vector in $\mathbb{R}^{n \times n}$ and let S be the set of all matrices that commute with A , that is,

$$S = \{B \mid AB = BA\}$$

Show that S is a subspace of $\mathbb{R}^{n \times n}$.

Question 13: (5 marks)

Let $\mathbf{x}_1, \dots, \mathbf{x}_k$ be linearly independent vectors in $\mathbb{R}^n$, and let A be a non-singular $n \times n$ matrix. Define $\mathbf{y}_i = A\mathbf{x}_i$, for $i = 1, 2, \dots, k$. Show that $\mathbf{y}_1, \dots, \mathbf{y}_k$ are linearly independent.

Question 14: (2 marks)

In $C[-\pi, \pi]$, find the dimension of the subspace spanned by $1, \cos(2x), \cos^2 x$.

Question 15: (3 marks)

Find the transition matrix corresponding to the change of coordinates on $\mathbb{R}^{2 \times 2}$, from the ordered basis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

to the ordered basis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Question 16: (3 marks)

Is it possible for an $m \times n$ to have a right inverse if $n < m$? Prove your answer mathematically.

Question 17: (3 marks)

Let S be the subspace of $C[a, b]$ spanned by e^x , xe^x , and x^2e^x . Let D be the differentiation operator of S . Find the matrix representing D with respect to $\{e^x, xe^x, x^2e^x\}$.

Question 18: (8 Marks)

Let V be the subspace of $C[a, b]$ spanned by $1, e^x, e^{-x}$, and let D be the differentiation operator on V .

(a) **2 Marks.**

Find the transition matrix S representing the change of coordinates from the ordered basis $[1, e^x, e^{-x}]$ to the ordered basis $[1, \frac{1}{2}(e^x + e^{-x}), \frac{1}{2}(e^x - e^{-x})]$.

(b) **2 Marks.**

Find the matrix A representing D with respect to the ordered basis $[1, \frac{1}{2}(e^x + e^{-x}), \frac{1}{2}(e^x - e^{-x})]$.

(c) **2 Marks.**

Find the matrix A representing D with respect to the ordered basis $[1, e^x, e^{-x}]$.

(d) **2 Mark.**

Verify that $B = S^{-1}AS$.

Question 19: (10 Marks)

Let A be an $m \times n$ matrix. Show that:

(a) **4 Marks**

If $\mathbf{x} \in N(A^T A)$, then $A\mathbf{x}$ is in both $R(A)$ and $N(A^T)$.

(b) **2 Marks**

$N(A^T A) = N(A)$.

(c) **2 Marks**

A and $A^T A$ have the same rank.

(d) **2 Marks**

If A has linearly independent columns, then $A^T A$ is nonsingular.

Question 20: (6 Marks)

Let $P = A(A^T A)^{-1}A^T$, where A is an $m \times n$ matrix of rank n .

(a) **2 Marks**

Show that $P^2 = P$.

(b) **2 Marks**

Prove that $P^k = P$ for $k = 1, 2, \dots$

(c) **2 Marks**

Show that P is symmetric.

Question 21: (2 Marks)

Let A be a 2×2 matrix. If $\text{tr}(A) = 8$, and $\det(A) = 12$, what are the eigenvalues of A ?

Question 22: (2 Marks)

Let A be a 4×4 matrix and let λ be an eigenvalue of algebraic multiplicity 3. If $A - \lambda I$ has rank 1, is A defective? Explain.

Question 23: (5 Marks)

Find the eigenvalues and the corresponding eigenspaces of the following matrix, and state the algebraic and geometric multiplicity of each eigenvalue.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Question 24: (2 Marks)

Let A be a diagonalizable matrix whose eigenvalues are all either 1 or -1. Show that $A^{-1} = A$.

Question 25: (3 Marks)

Show that any 3×3 matrix of the form

$$\begin{bmatrix} a & 1 & 0 \\ 0 & a & a \\ 0 & 0 & b \end{bmatrix}$$

is defective.